CUSTOMER SHOPPING BEHAVIOR

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Submitted To:

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01. INTRODUCTION

Customer shopping behaviour analysis is an important area in data analytics that helps businesses understand customer preferences, purchasing patterns, and buying habits. By analyzing shopping data, companies can identify trends and improve marketing strategies, customer satisfaction, and sales performance.

This project, titled “Customer Shopping Behaviour Analysis Using Python”, aims to analyze customer purchasing patterns using data analytics techniques and visualization tools. The dataset contains information such as customer age, gender, purchased items, category, purchase amount, season, payment method, shipping type, discount usage, and frequency of purchases.

Python programming and libraries such as Pandas, NumPy, Matplotlib, and Seaborn are used in this project for data cleaning, preprocessing, analysis, and visualization. Different charts and graphs are created to study relationships between variables and understand customer shopping trends.

The analysis helps in identifying seasonal purchasing behaviour, preferred product categories, spending habits, and customer preferences. These insights can support businesses in making better decisions related to product management, marketing strategies, and customer engagement.

Overall, this project demonstrates how data analytics and Python can be effectively used to analyze customer shopping behaviour and generate meaningful business insights.

02. DATASET DESCRIPTION

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
0	2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	36.0	Cash	Weekly
1	521	51	Male	Sunglasses	Accessories	84.0	South Carolina	M	White	Spring	3.9	Yes	Free Shipping	Yes	20.0	Debit Card	Quarterly
2	3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	18.0	Cash	Monthly
3	1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store Pickup	No	25.0	Cash	Annually
4	2929	40	Female	Jewelry	Accessories	80.0	Alabama	L	Yellow	Spring	3.6	No	Store Pickup	No	17.0	Credit Card	Weekly
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
5045	913	24	Male	Hoodie	Clothing	56.0	Arkansas	M	Green	Winter	3.4	Yes	Next Day Air	Yes	32.0	Bank Transfer	Monthly
5046	3466	19	Female	Backpack	Accessories	53.0	Iowa	M	Gold	Spring	3.4	No	Standard	No	31.0	Debit Card	Weekly
5047	3572	70	Female	Skirt	Clothing	27.0	North Dakota	M	White	Summer	3.2	No	Standard	No	15.0	Bank Transfer	Every 3 Months

Column Name	Description	Data Type
Customer ID	Unique ID of each customer.	int64
Age	Age of the customer.	int64
Gender	Gender of the customer.	object
Item Purchased	Product purchased by customer.	object
Category	Product category.	object
Purchase Amount (USD)	Amount spent on purchase.	float64
Location	Customer location/state.	object
Size	Size of the purchased item.	object
Color	Color of the purchased item.	object
Season	Season of purchase.	object
Review Rating	Customer rating for product.	float64
Subscription Status	Customer subscription availability.	object
Shipping Type	Type of delivery service.	object
Discount Applied	Whether discount was applied or not.	object
Previous Purchases	Number of previous purchases.	float64
Payment Method	Method used for payment.	object
Frequency of Purchases	How often customer makes purchases.	object

Dataset Source: Kaggle

Total Rows: 5050

Total Columns: 17

03. DATACLEANING AND PREPROCESSING

❖ IMPORTING DATA

The data was imported using pandas in python.

```
import pandas as pd
data=pd.read_csv("customer_shopping_behavior.csv")
data
```

❖ CHECK NULL VALUES

Checking missed values in the data.

```
data.isnull().any()
```

❖ CHECK SUM OF NULL VALUES

Checking sum of missed values in the data.

```
data.isnull().sum()
```

❖ CHECK DATA TYPES

Checking data types.

```
data.dtypes
```

❖ CHECK SKEWNESS

Checking skewness using kdeplot.

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.kdeplot(data['Purchase Amount (USD)'])
plt.ylabel("Density")
plt.xlabel("Amount")
plt.title("Purchase Amount Distribution")
plt.show()
```

❖ FILL NULL VALUES

Fill null values appropriately for each columns.

```
data['Purchase Amount (USD)']=data['Purchase Amount (USD)'].fillna(data['Purchase Amount (USD)'].median())
```

```
data['Size']=data['Size'].fillna("Unknown")
```

```
data['Review Rating']=data['Review Rating'].fillna(data['Review Rating'].median())
```

```
data['Previous Purchases']=data['Previous Purchases'].fillna(data['Previous Purchases'].median())
```

❖ CHECK DUPLICATES

Checking duplicated rows.

```
data.duplicated().any()
```

```
data.duplicated().sum()
```

```
data=data.drop_duplicates()
```

04. EXPLORATORY DATA ANALYIS(EDA)

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5050 entries, 0 to 5049
Data columns (total 17 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             5050 non-null   int64  
 1   Age                     5050 non-null   int64  
 2   Gender                  5050 non-null   object 
 3   Item Purchased          5050 non-null   object 
 4   Category                5050 non-null   object 
 5   Purchase Amount (USD)   4494 non-null   float64
 6   Location                5050 non-null   object 
 7   Size                    4680 non-null   object 
 8   Color                   5050 non-null   object 
 9   Season                  5050 non-null   object 
 10  Review Rating           4449 non-null   float64
 11  Subscription Status     5050 non-null   object 
 12  Shipping Type           5050 non-null   object 
 13  Discount Applied        5050 non-null   object 
 14  Previous Purchases      4502 non-null   float64
 15  Payment Method          5050 non-null   object 
 16  Frequency of Purchases  5050 non-null   object 
dtypes: float64(3), int64(2), object(12)
memory usage: 670.8+ KB
```

```
data.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	5050.000000	5050.000000	4494.000000	4449.000000	4502.000000
mean	2519.570891	44.150495	144.765236	3.668195	25.221901
std	1470.402964	15.282328	275.590101	0.865357	14.521635
min	1.000000	18.000000	10.120000	1.000000	0.000000
25%	1252.250000	31.000000	41.000000	3.000000	13.000000
50%	2499.500000	44.000000	65.000000	3.700000	25.000000
75%	3740.750000	57.000000	89.000000	4.400000	38.000000
max	5099.000000	70.000000	1499.760000	5.000000	50.000000

```
data.head()
```

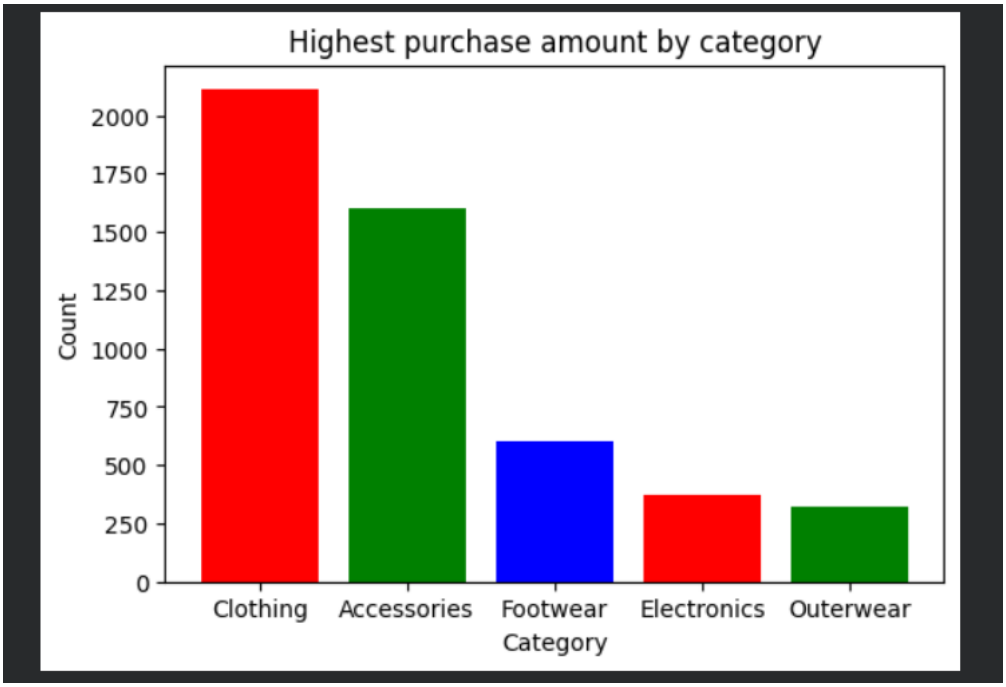
	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
0	2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	36.0	Cash	Weekly
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3	1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store Pickup	No	25.0	Cash	Annually
4	2929	40	Female	Jewelry	Accessories	80.0	Alabama	L	Yellow	Spring	3.6	No	Store Pickup	No	17.0	Credit Card	Weekly

```
data.tail()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
5045	913	24	Male	Hoodie	Clothing	56.0	Arkansas	M	Green	Winter	3.4	Yes	Next Day Air	Yes	32.0	Bank Transfer	Monthly
5046	3466	19	Female	Backpack	Accessories	53.0	Iowa	M	Gold	Spring	3.4	No	Standard	No	31.0	Debit Card	Weekly
5047	3572	70	Female	Skirt	Clothing	27.0	North Dakota	M	White	Summer	3.2	No	Standard	No	15.0	Bank Transfer	Every 3 Months
5048	3195	43	Female	Sneakers	Footwear	52.0	Maryland	L	Gold	Summer	4.6	No	Standard	No	10.0	Credit Card	Bi-Weekly
5049	1832	62	Male	Blouse	Clothing	71.0	Maine	M	Cyan	Summer	3.3	No	Free Shipping	No	5.0	Bank Transfer	Every 3 Months

1. BAR CHART

```
#Which category has the highest purchase amount
category_count=data['Category'].value_counts()
plt.figure(figsize=(6,4))
plt.bar(category_count.index,category_count.values,color=["red","green","blue"])
plt.title("Highest purchase amount by category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```

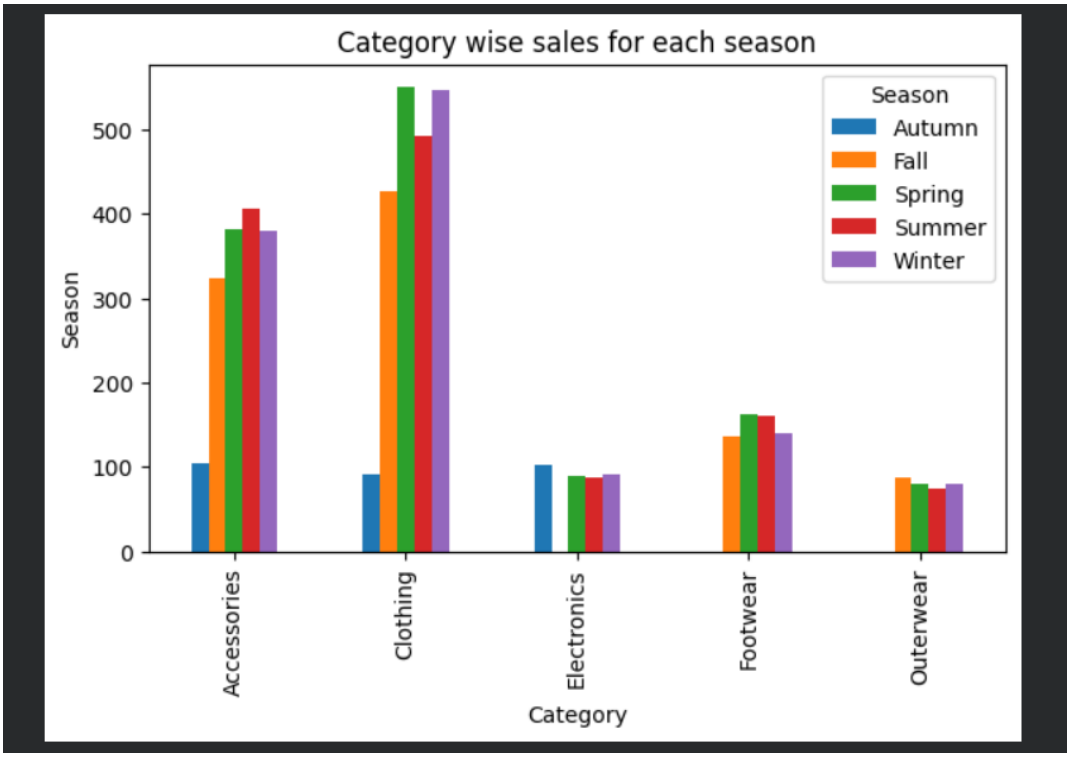


Inference:

The Clothing category has the highest number of purchases,followed by Accessories.Footwear,Electronics, and Outerwear have comparatively lower purchase counts.This indicates that customers purchase clothing items more frequently than other categories.

2.COLUMN CHART

```
#Category wise sales for each season
Season_sales=data.groupby(['Category','Season']).size().unstack()
Season_sales.plot(kind='bar',figsize=(7,4))
plt.title("Category wise sales for each season")
plt.xlabel("Category")
plt.ylabel("Season")
plt.show()
```

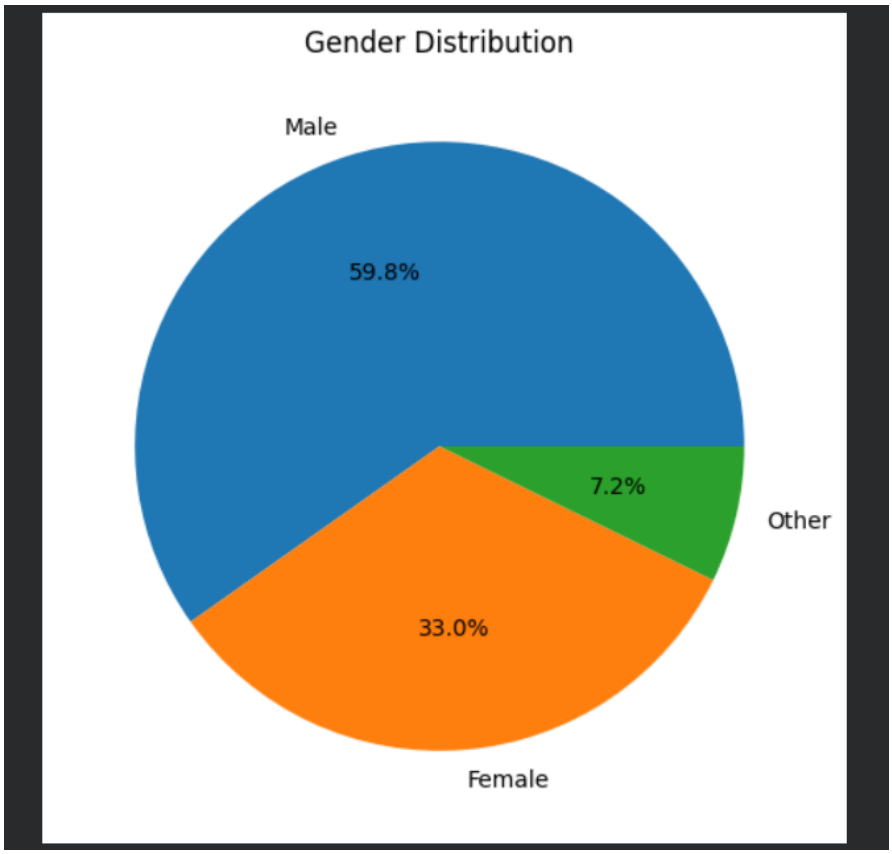


Inference:

The chart shows that clothing has the highest sales in every season.Accessories also have good sales, mainly in spring and summer.Footwear has medium sales,while electronics and outerwear have lower sales.Overall,sales are higher in spring,summer,and winter than in autumn.

3.PIE CHART

```
#Gender Distribution
gender_count=data['Gender'].value_counts()
plt.figure(figsize=(6,6))
plt.pie(gender_count,labels=gender_count.index,autopct='%1.1f%%')
plt.title("Gender Distribution")
plt.show()
```

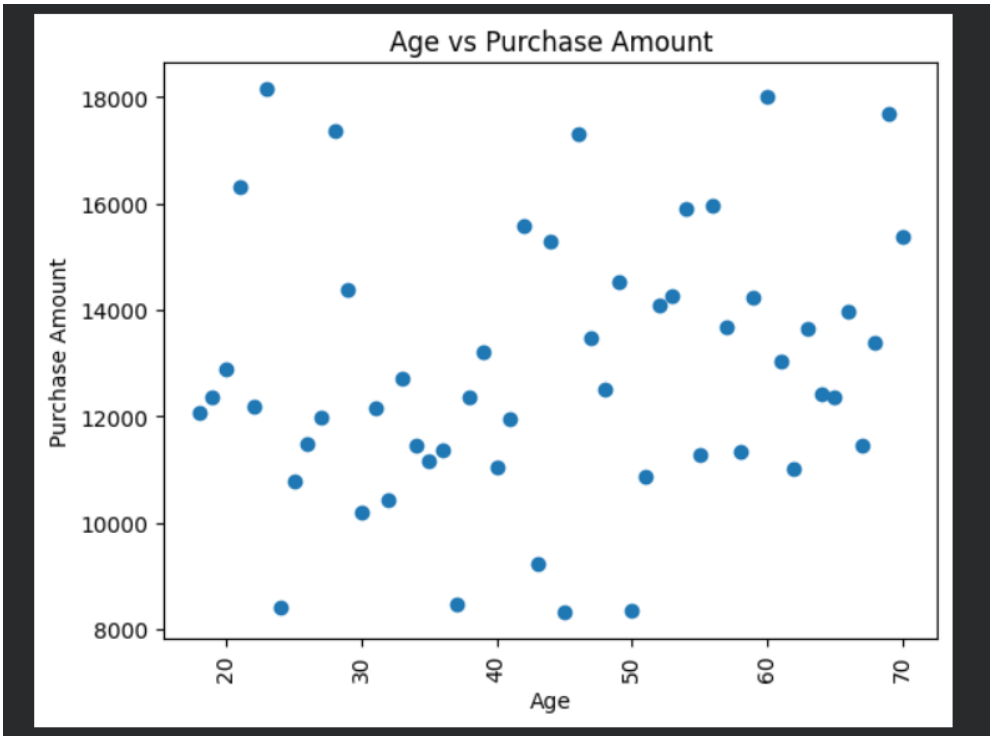


Inference:

The chart shows that males make up the majority of customers with 59.8%,followed by females at 33.0%,while the “Other” category accounts for 7.2%.This indicates that male customers contribute the largest share in the dataset.

4.SCATTER PLOT

```
#Age vs Purchase amount
Age_Purchase=data.groupby("Age")['Purchase Amount (USD)'].sum()
plt.scatter(Age_Purchase.index,Age_Purchase.values)
plt.title("Age vs Purchase Amount")
plt.xlabel("Age")
plt.ylabel("Purchase Amount")
plt.xticks(rotation=90)
plt.show()
```

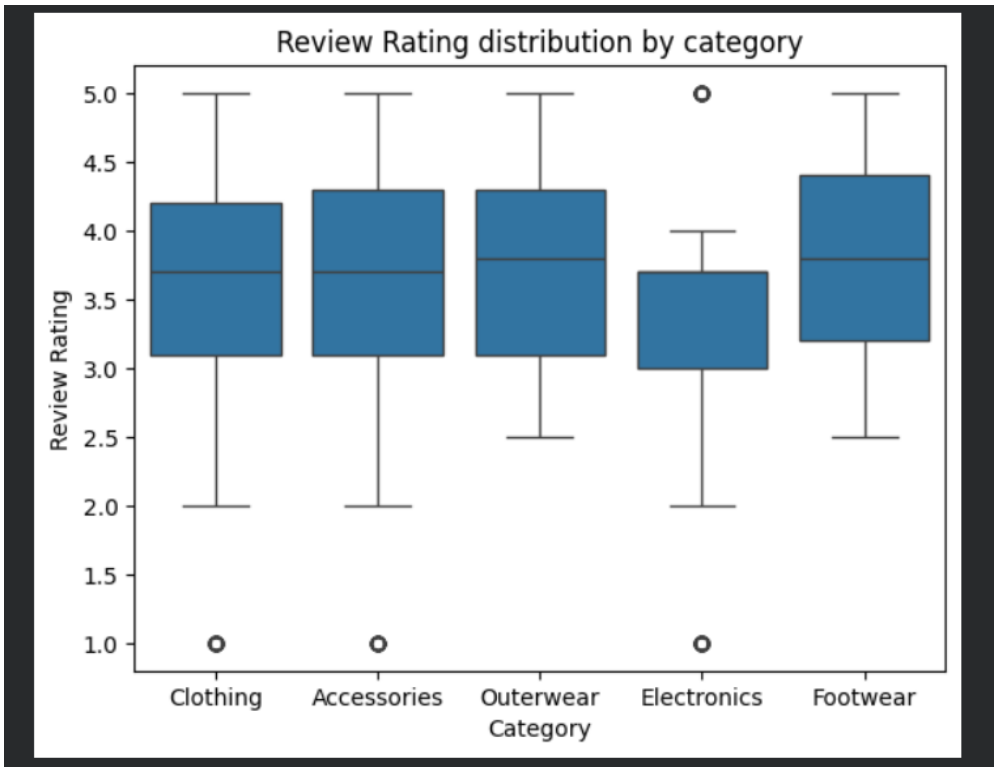


Inference:

The scatter plot shows that purchase amounts vary across different age groups.There is no clear increasing or decreasing trend between age and purchase amount,which means people of all ages make purchases at different spending levels.Most purchase amounts are between 10000 and 16000 USD.

5.BOX PLOT

```
#Review rating distribution by category
sns.boxplot(x='Category',y='Review Rating',data=data)
plt.title('Review Rating distribution by category')
plt.xlabel('Category')
plt.ylabel('Review Rating')
plt.show()
```

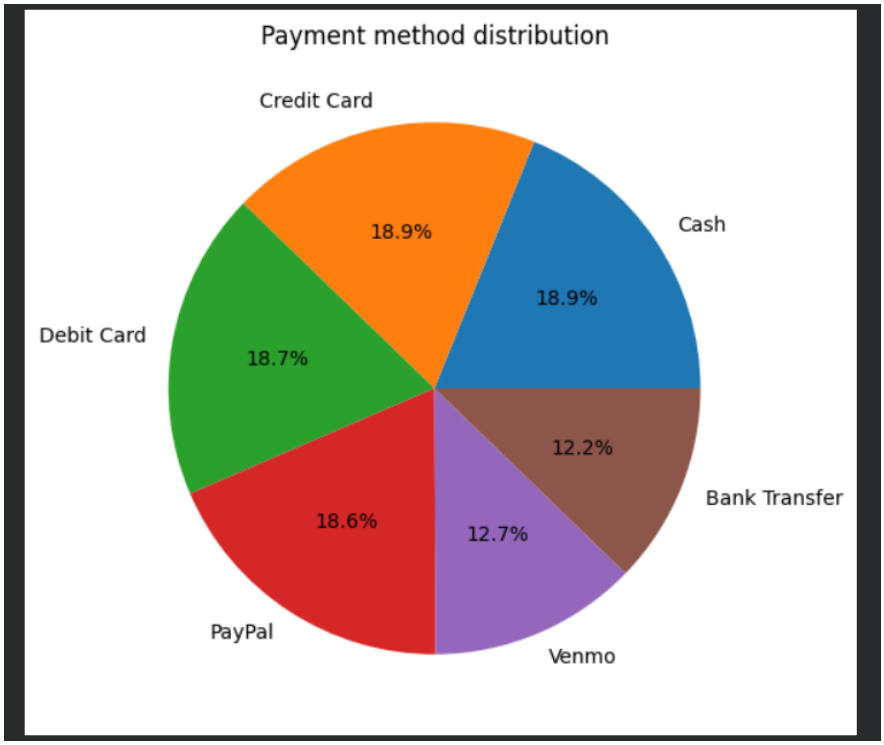


Inference:

The box plot shows that review ratings are fairly similar across all product categories.Most categories have average ratings between 3 and 4.5,indicating generally positive customer feedback.Footwear and Outerwear have slightly higher median ratings,while a few low-rated outliers are visible in Clothing,Accessories,and Electronics.

6.PIE CHART

```
#Which payment method is used the most
Payment_method=data['Payment Method'].value_counts()
plt.figure(figsize=(6,6))
plt.pie(Payment_method,labels=Payment_method.index,autopct='%1.1f%%')
plt.title("Payment method distribution")
plt.show()
```

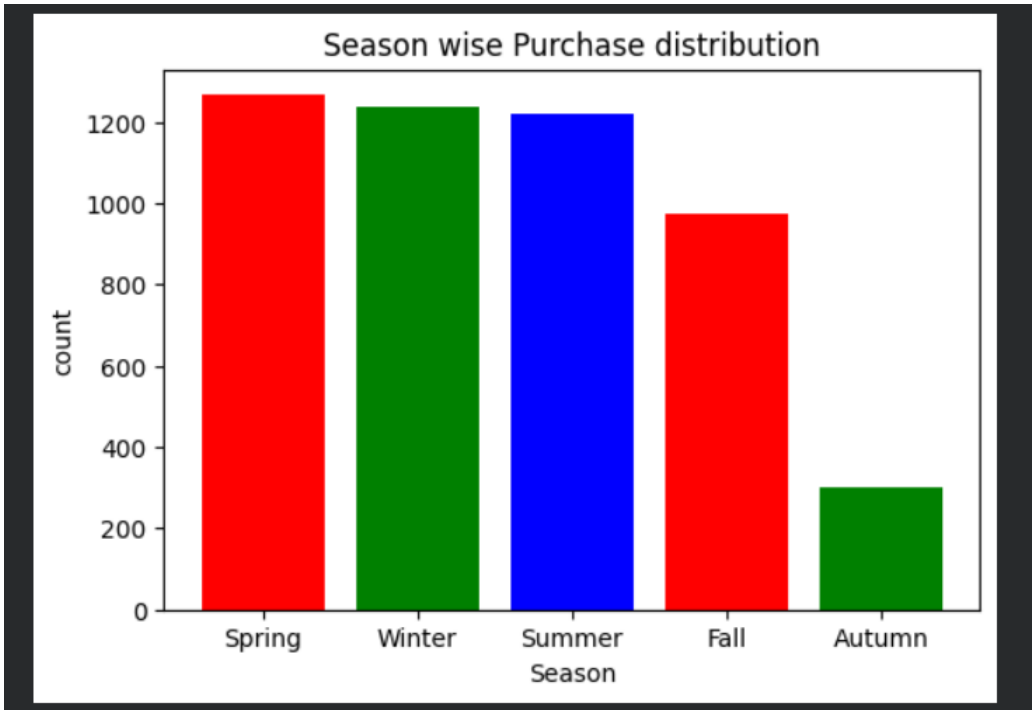


Inference:

The chart shows that Cash and Credit Card are the most used payment methods.Debit Card and PayPal are also used by many customers,while Venmo and Bank Transfer are used less compared to the others.

7.BAR CHART

```
#How does purchase amount vary across seasons
seasons_purchase=data["Season"].value_counts()
plt.figure(figsize=(6,4))
plt.bar(seasons_purchase.index,seasons_purchase.values,color=["red","green","blue"])
plt.title("Season wise Purchase distribution")
plt.xlabel("Season")
plt.ylabel("count")
plt.show()
```

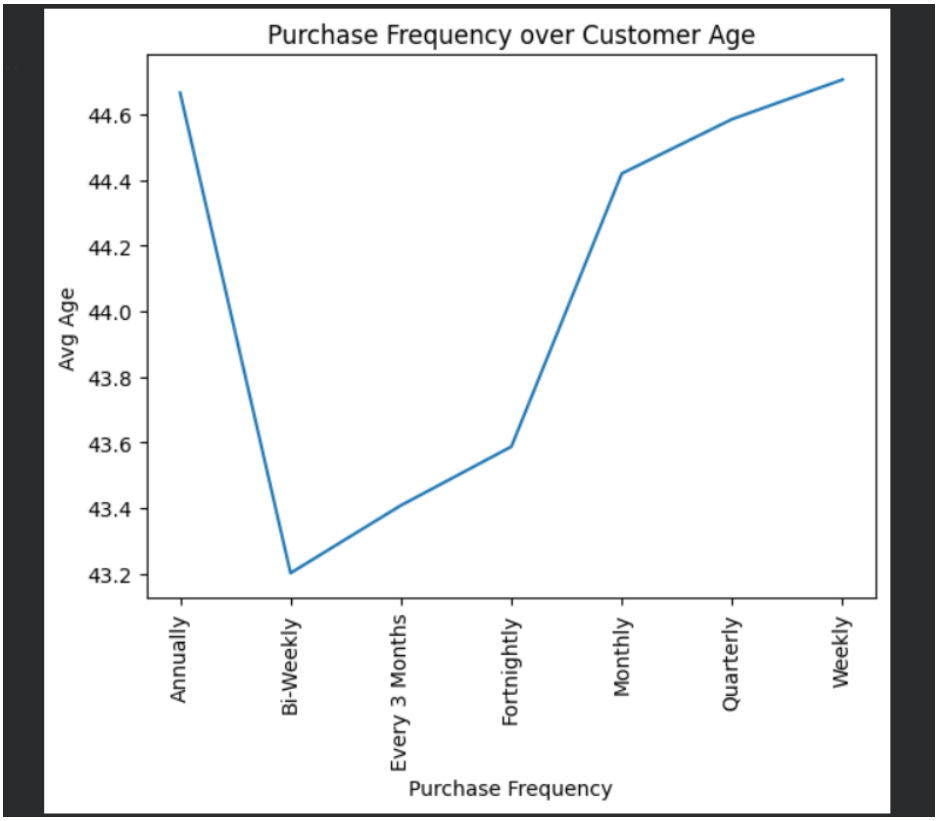


Inference:

The season-wise purchase distribution indicates that purchases are the highest during Spring, followed closely by Winter and Summer,showing strong customer demand in these seasons.Fall records a moderate number of purchases,while Autumn has the lowest purchase count by a significant margin.Overall,customer purchasing activity is concentrated in Spring,Winter,and Summer,with a noticeable decline during Fall and especially Autumn.

8.LINE CHART

```
#Frequency of purchases over customer age
Age_Purchase=data.groupby("Frequency of Purchases")["Age"].mean()
plt.plot(Age_Purchase.index,Age_Purchase.values)
plt.title("Purchase Frequency over Customer Age")
plt.xlabel("Purchase Frequency")
plt.ylabel("Avg Age")
plt.xticks(rotation=90)
plt.show()
```



Inference:

The graph shows that the average customer age changes slightly with different purchase frequencies.Customers who shop weekly,quarterly,and annually have a comparatively higher average age,while customers shopping bi-weekly have the lowest average age.Overall,the variation in age across purchase frequencies is small,indicating that purchase frequency is fairly similar among different age groups.

05. CONCLUSION

The analysis of the customer shopping behavior dataset provides valuable insights into customer preferences, purchasing patterns, and shopping trends. The study shows that factors such as age, gender, product category, seasonal demand, payment methods, discounts, and subscription status influence customer buying decisions. Customers with subscriptions and frequent purchase habits tend to shop more regularly, while discounts and convenient shipping methods also encourage higher engagement.

The dataset further highlights the importance of understanding customer demographics and preferences to improve business strategies. By analyzing purchase amounts, review ratings, and frequency of purchases, businesses can identify customer needs and enhance customer satisfaction. Overall, this study demonstrates how data analytics can help organizations make better marketing decisions, improve customer experience, and increase sales performance through data-driven insights.